

Mathematical Analysis II

Week 8 Homework (April 21)

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Section 10.1

1. 改变下列积分的顺序.

$$(2) \int_0^2 dx \int_{2x}^{6-x} f(x, y) dy;$$

解. $2x \leq y \leq 6 - x$, 所以 $x \leq \min\{6 - y, \frac{y}{2}\}$
所以原式 $= \int_0^4 dy \int_0^{\min\{6-y, \frac{y}{2}\}} f(x, y) dx$

$$(4) \int_a^b dy \int_y^b f(x, y) dx;$$

解. $y \leq x \leq b$, 所以 $y \leq x$, 所以原式 $= \int_a^b dx \int_a^x f(x, y) dy$

2. 计算下列积分.

$$(2) \iint_D \sin(x + y) dx dy, D = [0, \pi]^2;$$

解.

$$\begin{aligned} \text{原式} &= \int_0^\pi dx \int_0^\pi \sin(x + y) dy \\ &= \int_0^\pi \cos(x + y)|_0^\pi dx \\ &= \int_0^\pi (\cos(x + \pi) - \cos(x)) dx \\ &= -2 \int_0^\pi \cos x dx \\ &= 2 (\sin x)|_0^\pi \\ &= 0 \end{aligned}$$

$$(4) \iint_D (x + y) dx dy, D: \text{由 } x^2 + y^2 = a^2 \text{ 围成的圆在第一象限部分};$$

解. 令 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$, $0 \leq r \leq a$, $0 \leq \theta \leq \frac{\pi}{2}$, 所以 $dx dy = r dr d\theta$

$$\begin{aligned} \text{原式} &= \int_0^{\frac{\pi}{2}} d\theta \int_0^a r(\sin \theta + \cos \theta) r dr \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{\sin \theta + \cos \theta}{3} r^3 \right) \Big|_0^a d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{a^3}{3} (\sin \theta + \cos \theta) d\theta \\ &= \frac{a^3}{3} (-\cos \theta + \sin \theta) \Big|_0^{\frac{\pi}{2}} \\ &= \frac{2a^3}{3} \end{aligned}$$

(6) $\iint_D \frac{\sin y}{y} dx dy$, D : 由 $y = x$ 和 $x = y^2$ 围成;

解. 交点: $y = x = y^2$ 解得 $y = 0$ 或 $y = 1$

所以 $0 \leq y \leq 1$, $y^2 \leq x \leq y$

所以

$$\begin{aligned} \iint_D \frac{\sin y}{y} dx dy &= \int_0^1 dy \int_{y^2}^y \frac{\sin y}{y} dx \\ &= \int_0^1 (1 - y) \sin y dy \end{aligned}$$

令 $u = 1 - y$, $dv = \sin y dy$, 所以 $du = -1 dy$, $v = -\cos y$

所以

$$\begin{aligned} \int (1 - y) \sin y dy &= \int u dv = uv - \int v du \\ &= y \cos y - \cos y - \int \cos y dy \\ &= y \cos y - \cos y - \sin y + C \end{aligned}$$

所以原式 $= \int_0^1 (1 - y) \sin y dy = (y \cos y - \cos y - \sin y) \Big|_0^1 = 1 - \sin 1$

(8) $\iint_D |\cos(x + y)| dx dy$, 其中 D 是由直线 $y = x$, $y = 0$, $x = \frac{\pi}{2}$ 所围成;

解. 顶点: $(0, 0)$, $(\frac{\pi}{2}, 0)$, $(\frac{\pi}{2}, \frac{\pi}{2})$, 所以 $0 \leq x \leq \frac{\pi}{2}$, $0 \leq y \leq x$, 所以

$$\iint_D |\cos(x + y)| dx dy = \int_0^{\frac{\pi}{2}} dx \int_0^x |\cos(x + y)| dy$$

$$\text{其中 } \int_0^x |\cos(x+y)| dy = \begin{cases} \int_0^{\frac{\pi}{2}-x} \cos(x+y) dy - \int_{\frac{\pi}{2}-x}^x \cos(x+y) dy & \frac{\pi}{4} \leq x \leq \frac{\pi}{2} \\ \int_0^x \cos(x+y) dy & 0 \leq x < \frac{\pi}{4} \end{cases}$$

所以

$$\begin{aligned} & \iint_D |\cos(x+y)| dx dy \\ &= \int_0^{\frac{\pi}{4}} dx \int_0^x \cos(x+y) dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} dx \int_0^{\frac{\pi}{2}-x} \cos(x+y) dy - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} dx \int_{\frac{\pi}{2}-x}^x \cos(x+y) dy \\ &= \int_0^{\frac{\pi}{4}} (\sin 2x - \sin x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 - \sin x) dx - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin 2x - 1) dx \\ &= \left(-\frac{\cos 2x}{2} + \cos x \right) \Big|_0^{\frac{\pi}{4}} + (x + \cos x) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \left(\frac{\cos 2x}{2} + x \right) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= \frac{\pi}{2} - 1 \end{aligned}$$

3. 利用函数的奇偶性计算下列积分:

(1) $\iint_D (x^2 + y^2) dx dy$, $D: -1 \leq x \leq 1, -1 \leq y \leq 1$;

解. $x^2 + y^2$ 关于 x, y 均是偶函数, 所以

$$\begin{aligned} \iint_D (x^2 + y^2) dx dy &= 4 \int_0^1 dx \int_0^1 (x^2 + y^2) dy \\ &= 4 \int_0^1 \left(x^2 + \frac{1}{3} \right) dx \\ &= 4 \left(\frac{1}{3} + \frac{1}{3} \right) = \frac{8}{3} \end{aligned}$$

(2) $\iint_D \sin x \sin y dx dy$, $D: x^2 - y^2 = 1, x^2 + y^2 = 9$ 围成含原点的部分;

解. $\sin x \sin y$ 关于 x, y 均是奇函数, 且双曲线 $x^2 - y^2 = 1$, 圆 $x^2 + y^2 = 9$ 围成含原点的部分是关于关于原点中心对称的。

所以 $\iint_D \sin x \sin y dx dy = 0$

4. 设函数 φ 和 ψ 分别在区间 $[a, b]$ 和 $[c, d]$ 上可积, 求证 $f(x, y) = \varphi(x)\psi(y)$ 在 $D = [a, b] \times [c, d]$ 上可积, 且有

$$\iint_D f(x, y) dx dy = \int_a^b \varphi(x) dx \int_c^d \psi(y) dy.$$

证. 因为 φ 和 ψ 分别在区间 $[a, b]$ 和 $[c, d]$ 上可积, 所以任意 $a \leq x \leq b$, $f(x, \cdot) = \varphi(x)\psi$ 在 $[c, d]$ 上可积; 任意 $c \leq y \leq d$, $f(\cdot, y) = \varphi\psi(y)$ 在 $[a, b]$ 上可积。所以 $f = \varphi\psi$ 在 $[a, b] \times [c, d] = D$ 上可积

所以

$$\begin{aligned}\iint_D f(x, y) \, dx \, dy &= \iint_D \varphi(x) \psi(y) \, dx \, dy \\ &= \int_a^b \left(\int_c^d \varphi(x) \psi(y) \, dy \right) dx \\ &= \int_a^b \varphi(x) \left(\int_c^d \psi(y) \, dy \right) dx \\ &= \int_a^b \varphi(x) \, dx \int_c^d \psi(y) \, dy\end{aligned}$$

Q.E.D.

5. 设函数 $f(x)$ 在 $[0, a]$ 上连续, 证明

$$\begin{aligned}\int_0^a dx \int_0^x f(x) f(y) \, dy &= \frac{1}{2} \left(\int_0^a f(x) \, dx \right)^2, \\ \int_0^a dx \int_0^x f(y) \, dy &= \int_0^a (a-x) f(x) \, dx.\end{aligned}$$

证. 1. 因为 $0 \leq x \leq a$, $0 \leq y \leq x$, 所以 $0 \leq y \leq a$, $y \leq x \leq a$

$$\text{所以 } \int_0^a dx \int_0^x f(x) f(y) \, dy = \int_0^a dy \int_y^a f(x) f(y) \, dx = \int_0^a dx \int_x^a f(x) f(y) \, dy$$

所以

$$\begin{aligned}2 \int_0^a dx \int_0^x f(x) f(y) \, dy &= \int_0^a dx \int_0^x f(x) f(y) \, dy + \int_0^a dx \int_x^a f(x) f(y) \, dy \\ &= \int_0^a dx \int_0^a f(x) f(y) \, dy \\ &= \int_0^a f(x) \, dx \int_0^a f(y) \, dy \\ &= \left(\int_0^a f(x) \, dx \right)^2\end{aligned}$$

$$\text{所以 } \int_0^a dx \int_0^x f(x) f(y) \, dy = \frac{1}{2} \left(\int_0^a f(x) \, dx \right)^2$$

2.

$$\begin{aligned}\int_0^a dx \int_0^x f(y) \, dy &= \int_0^a dy \int_y^a f(y) \, dx \\ &= \int_0^a f(y) (a-y) \, dy \\ &= \int_0^a (a-x) f(x) \, dx\end{aligned}$$

Q.E.D.

6. 设函数 $f(x, y)$ 有连续的二阶偏导数, 在 $D = [a, b] \times [c, d]$ 上, 求积分

$$\iint_D \frac{\partial^2 f(x, y)}{\partial x \partial y} dx dy.$$

解.

$$\begin{aligned} \iint_D \frac{\partial^2 f(x, y)}{\partial x \partial y} dx dy &= \int_a^b dx \int_c^d \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}(x, y) \right) dy \\ &= \int_a^b \left(\frac{\partial f}{\partial x}(x, d) - \frac{\partial f}{\partial x}(x, c) \right) dx \\ &= f(b, d) - f(a, d) - f(b, c) + f(a, c) \end{aligned}$$

7. 设函数 $f(x, y)$ 连续, 求极限

$$\lim_{r \rightarrow 0} \frac{1}{\pi r^2} \iint_{x^2 + y^2 \leq r^2} f(x, y) dx dy.$$

解. $D = \{(x, y) : x^2 + y^2 \leq r^2\}$, $\sigma(D) = \pi r^2$

所以存在 $P_r(x_r, y_r) \in D$ s.t. $f(P_r)\sigma(D) = \iint_D f(x, y) dx dy$

所以 $\frac{1}{\pi r^2} \iint_D f(x, y) dx dy = f(P_r)$

当 $r \rightarrow 0$, $P_r \rightarrow (0, 0)$, 所以 $\frac{1}{\pi r^2} \iint_D f(x, y) dx dy \rightarrow f(0, 0)$